

A Hierarchical Mass-Energy Functional for Localized Vortex Structures: Kinematic Scale, Geometric Shielding, and Topological State Classification

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Abstract

We derive and organise a hierarchical mass-energy functional $M_K = \rho_m r_c^3 \Pi_K \Xi_K S_t^{-2}$ for a localised vortex filament in an ideal incompressible fluid, where $\rho_m = \frac{1}{2}\rho_f v^2/c^2$ is the kinetic mass-equivalent density, r_c is the vortex core radius, Π_K is a dimensionless geometric shielding factor, Ξ_K a topological state kernel, and S_t a local time-dilation factor. Four results are established without circular reasoning. First, the kinetic energy density $\rho_E = \frac{1}{2}\rho_f v^2$ and its mass-equivalent $\rho_m = \rho_E/c^2$ follow from standard fluid mechanics and special relativity. Second, for a vortex filament whose core radius satisfies $r_c = \alpha\hbar/(2m_e c)$ (the Compton-scale core hypothesis), the dimensionless length ratio $\lambda_c/(\pi r_c) = 4/\alpha$ follows by pure algebra from the definitions of the Compton wavelength $\lambda_c = h/(m_e c)$ and the fine-structure constant $\alpha = e^2/(4\pi\epsilon_0\hbar c)$; no additional assumptions are required. Third, the local time-dilation factor $S_t = (1 - v^2/c^2)^{1/2}$ is the standard special-relativistic expression applied to the circulation speed. Fourth, the topological kernel Ξ_K is explicitly identified as an open modelling ansatz whose coefficient values are not yet derived from first principles; the paper proposes a concrete parameterisation and lists the measurable consequences that would confirm or refute it. The functional is proposed as an organising framework for further analytical and numerical work.

Keywords: relational time, time of arrival, quantum time observables, covariant observables, event currents, clock holonomy, quantum field theory

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1 Introduction and Scope

The mass-energy of a localised fluid structure is determined by its internal kinetic energy, the spatial extent of the structure, any geometric factors that modulate the coupling to the surrounding medium, and the identity of the topological state. These four contributions have been studied separately in the literature on vortex dynamics [1, 2], topological field theory [8], and analogue gravity [11], but have not been assembled into a single hierarchical formula with explicit status labels for each factor.

The purpose of this paper is to do exactly that. We derive each factor in turn, assign it an unambiguous epistemic label (derived, calibrated, or conjectural), and assemble the factors into a single mass-energy functional. The result is an organising framework rather than a predictive theory: it is predictive only when the topological kernel Ξ_K is specified by an independent theory of the vortex topology.

We work throughout in SI units. Greek indices μ, ν run over spacetime; Latin indices i, j, k over spatial coordinates. The speed of light in vacuo is c ; $\hbar = h/(2\pi)$; m_e and e are the electron mass and charge; ε_0 is the permittivity of free space.

2 Kinematic Scale and Mass-Equivalent Density

2.1 Kinetic energy density of a circulatory flow

Consider an ideal incompressible fluid of mass density ρ_f (kg m^{-3}) supporting a localised circulatory velocity field $\mathbf{v}(\mathbf{x})$. The local kinetic energy density is [1, 3]

$$\rho_E(\mathbf{x}) = \frac{1}{2}\rho_f |\mathbf{v}(\mathbf{x})|^2. \quad (1)$$

For a vortex whose peak circulation speed is v , the characteristic energy density is $\rho_E = \frac{1}{2}\rho_f v^2$.

2.2 Mass-equivalent density via $E = mc^2$

By the special-relativistic energy–mass equivalence [4], any energy density ρ_E contributes a mass-equivalent density

$$\rho_m = \frac{\rho_E}{c^2} = \frac{\rho_f v^2}{2c^2}. \quad (2)$$

This expression is dimensionally consistent: $[\rho_f v^2/c^2] = \text{kg m}^{-3}$. No assumptions beyond standard fluid mechanics and special relativity are used.

2.3 Bare mass scale

The mass enclosed in a spherical core of radius r_c is

$$M_\star = \rho_m r_c^3 = \frac{\rho_f v^2}{2c^2} r_c^3. \quad (3)$$

Here r_c is a free parameter (the vortex core radius) until it is specified by the topological model; see section 3 and section 8.

3 Geometric Shielding Gate

3.1 Compton-scale core hypothesis

Definition 1 (Compton-scale core). *We adopt as a modelling hypothesis the identification*

$$r_c = \frac{\alpha \hbar}{2m_e c}, \quad (4)$$

where $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ is the fine-structure constant. This is a calibration, not a derivation: no microscopic mechanism is asserted here, and the hypothesis is testable by any experiment that independently constrains the vortex core radius.

3.2 Algebraic derivation of the shielding ratio

Lemma 1 (Shielding ratio). *Under definition 1, the ratio of the electron Compton wavelength $\lambda_c = h/(m_e c)$ to the quantity πr_c satisfies*

$$\frac{\lambda_c}{\pi r_c} = \frac{4}{\alpha}. \quad (5)$$

Proof. The Compton wavelength is $\lambda_c = h/(m_e c) = 2\pi\hbar/(m_e c)$. Substituting eq. (4):

$$\frac{\lambda_c}{\pi r_c} = \frac{2\pi\hbar/(m_e c)}{\pi \cdot \alpha\hbar/(2m_e c)} = \frac{2\pi\hbar}{m_e c} \cdot \frac{2m_e c}{\pi\alpha\hbar} = \frac{4}{\alpha}. \quad (6)$$

The result is an algebraic identity among λ_c , r_c , and α . It does not depend on the numerical value of any of these quantities, and introduces no new assumptions beyond definition 1. $\square$

Remark 1. *The ratio $4/\alpha \approx 548.1$ is numerically large because $\alpha \approx 1/137.036$. This amplification factor is therefore a direct consequence of the smallness of the fine-structure constant and the assumption that the core radius is set at the Compton scale.*

3.3 Geometric shielding factor

We introduce a binary exposure index $G(K) \in \{0, 1\}$ for a topological state K (see section 5) and define the geometric shielding factor

$$\Pi_K = \left(\frac{\lambda_c}{\pi r_c} \right)^{G(K)} = \left(\frac{4}{\alpha} \right)^{G(K)}. \quad (7)$$

When $G(K) = 0$ the factor is unity (no geometric amplification); when $G(K) = 1$ the factor is $4/\alpha \approx 548$. The physical interpretation is that states with $G(K) = 1$ couple to the medium at the Compton scale, while states with $G(K) = 0$ remain in the bare core regime. The assignment of $G(K)$ to specific topological states is a *modelling choice* that lies outside the scope of this paper.

4 Local Time-Dilation Factor

A fluid element moving with speed $v = |\mathbf{v}|$ relative to the laboratory frame experiences a standard special-relativistic time dilation [12, 13]:

$$S_t = \sqrt{1 - \frac{v^2}{c^2}}, \quad (8)$$

with $d\tau = S_t dt$. For $v \ll c$ one has $S_t \approx 1 - v^2/(2c^2)$ and $S_t^{-2} \approx 1 + v^2/c^2$. This factor appears in the mass-energy functional through the local effective inertia: a region of strong circulation is both kinetically energetic and locally time-dilated; the combination amplifies the effective mass.

5 Topological State Kernel

5.1 Motivation and status

The topological identity of a vortex filament is encoded in its knot type K . In the mathematical theory of knotted vortices, three families of invariants are relevant [5, 7, 9]:

- (i) *Crossing invariants* $C(K)$, such as the crossing number or writhe, which count the complexity of the knot diagram;
- (ii) *Geometric invariants* $L(K)$, such as the ropelength [10], which encode the minimum length required to realise the knot at a given core thickness;
- (iii) *Helicity invariants* $\mathcal{H}(K)$, related to the Gauss linking integral and the Hopf invariant [5, 6], which measure the self-linking of the vorticity field.

A natural dimensionless state kernel is a linear combination of these invariants.

5.2 Proposed kernel ansatz

Definition 2 (Topological kernel). *We propose the ansatz*

$$\Xi_K = [\alpha_C C(K) + \beta_L L(K) + \gamma_H \mathcal{H}(K)] \varphi^{-2k(K)}, \quad (9)$$

where $\alpha_C, \beta_L, \gamma_H$ are dimensionless coefficients, $k(K) \in \mathbb{Z}_{\geq 0}$ is a discrete state-level index, and $\varphi = (1 + \sqrt{5})/2$ is the golden ratio.

Remark 2 (Epistemic status of definition 2). *The kernel (9) is a conjectural modelling ansatz, not a derivation. Three elements are not justified from first principles in this paper:*

- (i) *The coefficients $\alpha_C, \beta_L, \gamma_H$ are free parameters; their values must be determined by fitting to observed particle masses or by a separate dynamical derivation.*
- (ii) *The golden-ratio ladder φ^{-2k} is motivated by the fixed-point structure of the nonlinear Schrödinger equation on a closed loop (cf. [14, 15]) but is not derived from a variational principle for the present system.*
- (iii) *The binary exposure index $G(K)$ is assigned to knot types by a rule not stated in this paper.*

These three open points define the research programme that this paper proposes.

6 The Hierarchical Mass-Energy Functional

6.1 Assembly

Combining eqs. (3) and (7) to (9), we define

Proposition 1 (Hierarchical mass-energy functional). *Under the kinematic, Compton-core, time-dilation, and topological-kernel hypotheses of sections 2 to 5, the mass-energy of a topological vortex state K at position $\mathbf{x}$ is*

$$M_K(\mathbf{x}) = \frac{\rho_f(\mathbf{x}) |\mathbf{v}(\mathbf{x})|^2}{2c^2} r_c^3 \left(\frac{4}{\alpha} \right)^{G(K)} \Xi_K \left(1 - \frac{|\mathbf{v}(\mathbf{x})|^2}{c^2} \right)^{-1} \quad (10)$$

with Ξ_K as in definition 2.

Proof. Substitute eq. (2) into eq. (3), apply eq. (7), include the state kernel definition 2, and note that $S_t^{-2} = (1 - v^2/c^2)^{-1}$. The result follows by direct substitution. The dimensional check gives $[\rho_f v^2 / c^2 \cdot r_c^3] = \text{kg m}^{-3} \text{m}^3 = \text{kg}$; all remaining factors are dimensionless. $\square$

6.2 Factored interpretation

The functional separates into four physically distinct sectors:

$$M_K = \underbrace{\rho_m r_c^3}_{\text{medium scale}} \times \underbrace{(4/\alpha)^{G(K)}}_{\text{geometric gate}} \times \underbrace{\Xi_K}_{\text{topological kernel}} \times \underbrace{S_t^{-2}}_{\text{clock factor}}. \quad (11)$$

Each factor is independently testable in principle: the medium scale is set by macroscopic fluid measurements; the geometric gate follows from the core-radius hypothesis; the topological kernel depends only on knot invariants; the clock factor is a standard relativistic correction.

7 Reduction Limits

7.1 Static limit ($v \ll c$)

For $v \ll c$, $S_t^{-2} \rightarrow 1$ and eq. (10) reduces to

$$M_K \approx \rho_m r_c^3 (4/\alpha)^{G(K)} \Xi_K, \quad (12)$$

the static mass spectrum.

7.2 Homogeneous medium

If ρ_f and v are spatially uniform, ρ_m is constant and the spatial integration over the core volume gives a factor $\propto r_c^3$, consistent with eq. (3).

7.3 Weak-circulation expansion

For $v^2/c^2 \ll 1$,

$$S_t^{-2} = 1 + \frac{v^2}{c^2} + \mathcal{O}\left(\frac{v^4}{c^4}\right), \quad (13)$$

so the leading relativistic correction to the mass is positive and quadratic in the circulation speed.

8 Calibration and Numerical Check

8.1 Calibrated parameters

The functional eq. (10) contains three parameters that are not derived in this paper and must be supplied externally:

- (i) ρ_f : the background effective fluid density. We adopt the phenomenological value $\rho_f = 7.0 \times 10^{-7} \text{ kg m}^{-3}$, consistent with calibrations of the electron vortex model against the electron mass [18]. This value is a *calibration*, not a prediction.
- (ii) v : the characteristic circulation speed. We adopt $v = \alpha c/2 \approx 1.094 \times 10^6 \text{ m s}^{-1}$, the value that identifies the circulation speed with $\alpha c/2$ [18]. Again, this is a *calibration*.
- (iii) r_c : by definition 1, $r_c = \alpha \hbar / (2m_e c) \approx 1.409 \times 10^{-15} \text{ m}$.

The topological coefficients α_C , β_L , γ_H and the level index $k(K)$ are open parameters to be determined.

8.2 Numerical benchmark

With the calibrated values above:

$$\rho_E = \frac{1}{2}\rho_f v^2 \approx 4.188 \times 10^5 \text{ J m}^{-3}, \quad (14)$$

$$\rho_m = \rho_E/c^2 \approx 4.659 \times 10^{-12} \text{ kg m}^{-3}, \quad (15)$$

$$M_\star = \rho_m r_c^3 \approx 1.303 \times 10^{-56} \text{ kg}. \quad (16)$$

The geometric gate evaluates to

$$\frac{4}{\alpha} \approx 548.14, \quad (17)$$

in exact numerical agreement with $\lambda_c/(\pi r_c)$, which confirms lemma 1 numerically. The local time-dilation factor at the calibrated speed is

$$S_t = \sqrt{1 - v^2/c^2} \approx 0.999993, \quad S_t^{-2} \approx 1.000013, \quad (18)$$

so the clock correction is at the level of 1.3×10^{-5} and is negligible in the present calibration regime.

The bare scale $M_\star \approx 1.3 \times 10^{-56} \text{ kg}$ is much smaller than the electron mass $m_e \approx 9.1 \times 10^{-31} \text{ kg}$. The ratio $m_e/M_\star \approx 7 \times 10^{25}$ must be supplied by the product $\Pi_K \Xi_K$, which is the open problem that the topological kernel definition 2 is designed to address.

9 Falsifiable Consequences

The functional eq. (10) yields three testable predictions that are independent of the open parameters $\alpha_C, \beta_L, \gamma_H$:

- (i) **Scaling with core radius.** Equation (3) predicts $M_\star \propto r_c^3$ for fixed fluid parameters. An independent measurement of r_c that is inconsistent with $r_c = \alpha\hbar/(2m_e c)$ would falsify definition 1.
- (ii) **Binary geometric clustering.** States with $G(K) = 1$ should have masses larger than states with $G(K) = 0$ by a factor of order $4/\alpha \approx 548$, holding all other factors equal. This is a sharp prediction that does not require knowledge of Ξ_K .
- (iii) **Mass-clock correlation.** If the same circulation field $\mathbf{v}(\mathbf{x})$ is responsible for both the inertial mass and the time dilation, regions of higher effective mass should exhibit a proportionally larger proper-time retardation. The coupling is $\delta M/M \approx v^2/c^2$ and $\delta\tau/\tau \approx -v^2/(2c^2)$ at leading order, giving

$$\frac{\delta M}{M} \approx -2 \frac{\delta\tau}{\tau}. \quad (19)$$

This relation is independent of all calibrated parameters.

10 Status Summary

Factor	Status	Dependencies
$\rho_E = \frac{1}{2}\rho_f v^2$	Derived (fluid mechanics)	Batchelor [1], Lamb [3]
$\rho_m = \rho_E/c^2$	Derived ($E = mc^2$)	Einstein [4]
$\lambda_c/(\pi r_c) = 4/\alpha$	Derived (algebra, given definition 1)	lemma 1
$S_t = (1 - v^2/c^2)^{1/2}$	Derived (special relativity)	Will [12]
$r_c = \alpha\hbar/(2m_e c)$	Calibration hypothesis	definition 1
$\rho_f = 7.0 \times 10^{-7} \text{ kg m}^{-3}$	Calibrated	section 8.1
$v = \alpha c/2$	Calibrated	section 8.1
$G(K) \in \{0, 1\}$	Modelling rule, not derived	Open
$\alpha_C, \beta_L, \gamma_H$	Free parameters	Open
$k(K)$ (golden-ladder level)	Conjectural ansatz	Open

11 Discussion and Outlook

The functional eq. (10) separates cleanly into established physics (kinetic energy density, mass–energy equivalence, special-relativistic time dilation, algebraic properties of the fine-structure constant) and open modelling choices (core-radius hypothesis, topological kernel ansatz, golden-ladder level assignment). The boundary between the two is made explicit in section 10.

The main open problem is the determination of Ξ_K from a microscopic or variational principle. Partial progress has been made using the knotted-vortex energy functionals of Faddeev and Niemi [8] and the ropelength bounds of Cantarella et al. [10]; a complete derivation remains outstanding.

The core-radius hypothesis definition 1 could in principle be tested via high-precision measurements of the electron charge radius or by atomic spectroscopy experiments sensitive to the inner structure of the Coulomb potential at the femtometre scale. Existing Lamb-shift data constrain any additional short-range contribution to the potential at a level that is broadly consistent with, but does not uniquely select, the value $r_c = \alpha\hbar/(2m_e c)$ [16, 17].

This paper employs the functional as a toy model for the purpose of identifying which parts of a localized-vortex mass formula are generic (following from fluid mechanics and special relativity) and which parts require a specific microscopic model. The three falsifiable predictions of section 9 hold for any model that shares the kinematic and geometric sectors, irrespective of the topological kernel.

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